NAME: _____ Geometry (FL B.E.S.T.)

10.5 Solids of Revolution

Spinning a flat 2D region a full $360°$ around a line (the **axis of revolution**) sweeps out a 3D **solid of revolution**. The three building blocks are: a rectangle about one of its sides sweeps a **cylinder**; a right triangle about a leg sweeps a **cone**; and a semicircle about its diameter sweeps a **sphere**. More advanced regions build more advanced solids: a right trapezoid about its perpendicular leg sweeps a **frustum** (a cone with the tip cut off), a rectangle that does *not* touch the axis sweeps a **hollow cylinder (tube)** — found by subtracting an inner cylinder from an outer one — and a region offset entirely from the axis sweeps a **torus** (donut shape). This worksheet identifies the solid *and* computes its volume.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Square side 7 rotated about one side. Identify the solid and find its volume.

2. Legs 5 and 12; rotate about leg 12. Identify the solid and find its volume.

3. Semicircle $r = 6$ about its diameter. Identify the solid and find its volume.

4. 3×8 rectangle rotated about the side of length 3. Find the volume.

5. The same 3×8 rectangle rotated about the side of length 8 instead. Find the volume.

6. Right trapezoid, parallel sides 6 and 2, perpendicular leg 9, rotated about that leg. Find the volume.

7. A rectangle spans from 2 to 6 units from the axis, height 12, rotated $360°$. Find the volume.

8. A cylinder ($r = 3$, $h = 5$) is topped by a cone ($r = 3$, $h = 4$) on the same axis. Find the total volume.

9. Rectangle rotated about one side. Name the solid.

10. Right triangle rotated about a leg. Name the solid.

11. Semicircle rotated about its diameter. Name the solid.

12. Right trapezoid rotated about its perpendicular leg. Name the solid.

NAME: _____ Geometry (FL B.E.S.T.)

13. Rectangle rotated with a gap between it and the axis. Name the solid.

14. A vase's outer profile is modeled as a frustum: top radius 3, bottom radius 6, height 10. Find its volume.

15. A right triangle with legs 9 and 12 (hypotenuse 15) is rotated about its hypotenuse. Find the volume of the resulting bicone.

✓ Answer Key

1. Cylinder; $V = 343\pi$
2. Cone; $V = 100\pi$
3. Sphere; $V = 288\pi$
4. 192π
5. 72π
6. 156π
7. 384π
8. 57π
9. Cylinder
10. Cone
11. Sphere
12. Frustum
13. Hollow cylinder (tube)
14. 210π
15. $\frac{1296\pi}{5}$ (=259.2π)